4. Using solvent extraction to remove one component.
5. By freezing one of the components.
6. By applying centrifugal force.
7. Adding dehydrating agents for water in oil (W/O) type.
8. Using ultrasonic waves.
9. Heating at high pressures.

Inversion of Phase:

The change of W/O emulsion into O/W emulsion is called inversion of phases.

For example:

An oil in water emulsion containing potassium soap as emulsifying agent can be converted into water in oil emulsion by adding $CaCl_2$ or $AlCl_3$. The mechanism of inversion is in the recent developments of research.

10.7 Various application of colloids

In every path of life, colloids play a great role. Human body contains the numerous colloidal solutions. The blood in our body, protoplasma of plant and animal cell, and fats in our intestines are in the form of emulsions. Synthetic polymers like polystyrene silicones and PVC are colloids.

Food

Food stuffs like milk cream, butter, etc are present in colloidal form.

Medicines

Antibodies such as penicillin and streptomycin are produced in colloidal form for suitable injections. Colloidal gold and colloidal calcium are used as tonics. Milk of magnesia is used for stomach troubles. Silver sol protected by gelatine known as Argyrol is used as eye lotion.

In Industry

Colloids find many applications in industries.

(i) Water purification:

Purification of drinking water is activated by coagulation of suspended impurities in water using alums containing Al^{3+}

(ii) In washing:

The cleansing action of soap is due to the formation of emulsion of soap molecules with dirt and grease.

(iii) Tanning of leather

Skin and hides are protein containing positively charged particles which are coagulated by adding tannin to give hardened leather for further application. Chromium salts are used for the purpose. Chrome tanning can produce soft and polishable leather.

XII U10-Surface Chemistry.indd 98 11-01-2021 15:33:48